



June 12, 2026

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Effluent PFAS
Pace Project No.: 50431361

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on May 14, 2026. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Minneapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,

A handwritten signature in blue ink that reads "Brian Hall".

Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures

cc:



REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: Effluent PFAS

Pace Project No.: 50431361

Pace Analytical Services, LLC - Minneapolis MN

1700 Elm Street SE, Minneapolis, MN 55414

Alaska Contaminated Sites Certification #: 17-009

Arizona Certification #: AZ0014

Arkansas WW Certification #: 88-0680

Colorado Certification #: MN00064

DoD via A2LA Certification #: 2926.01

Florida Certification #: E87605

Idaho Certification #: MN00064

Indiana Certification #: C-MN-01

ISO/IEC 17025 via A2LA Certification #: 2926.01

Kentucky DW Certification #: 90062

Louisiana DEQ Certification #: AI-03086

Maine Certification #: MN00064

Michigan Certification #: 9909

Minnesota Dept of Ag Approval via Certification #: 027-053-137

Mississippi Certification #: MN00064

Montana Certification #: CERT0092

Nevada Certification #: MN00064

New Jersey Certification #: MN002

North Carolina DW Certification #: 27700

North Dakota Certification via A2LA #: R-036

Ohio DW Certification #: 41244

Oklahoma Certification #: 9507

Oregon Secondary Certification #: MN200001

Puerto Rico Certification #: MN00064

Tennessee Certification #: TN02818

Utah Certification #: MN00064

Virginia Certification #: 460163

West Virginia DEP Certification #: 382

Wisconsin Certification #: 999407970

USDA Permit #: P330-19-00208

Alabama Certification #: 40770

Alaska DW Certification #: MN00064

Arkansas DW Certification #: MN00064

California Certification #: 2929

Connecticut Certification #: PH-0256

EPA Region 8 Tribal Water Systems+Wyoming DW Certification #: via MN 027-053-137

Georgia Certification #: 959

Illinois Certification #: 200011

Iowa Certification #: 368

Kansas Certification #: E-10167

Kentucky WW Certification #: 90062

Louisiana DW Certification #: MN00064

Maryland Certification #: 322

Minnesota Certification #: 027-053-137

Minnesota Petrofund Registration #: 1240

Missouri Certification #: 10100

Nebraska Certification #: NE-OS-18-06

New Hampshire Certification #: 2081

New York Certification #: 11647

North Carolina WW Certification #: 530

North Dakota Certification via MN #: R-036

Ohio VAP Certification (1700) #: CL101

Oregon Primary Certification #: MN300001

Pennsylvania Certification #: 68-00563

South Carolina Certification #: 74003001

Texas Certification #: T104704192

Vermont Certification #: VT-027053137

Washington Certification #: C486

West Virginia DW Certification #: 9952 C

Wyoming UST Certification via A2LA #: 2926.01

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SAMPLE SUMMARY

Project: Effluent PFAS
Pace Project No.: 50431361

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50431361001	Effluent	Water	05/13/26 11:22	05/14/26 10:15

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SAMPLE ANALYTE COUNT

Project: Effluent PFAS
Pace Project No.: 50431361

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50431361001	Effluent	EPA 1633	NBH	65	PASI-M

PASI-M = Pace Analytical Services - Minneapolis

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SUMMARY OF DETECTION

Project: Effluent PFAS

Pace Project No.: 50431361

Lab Sample ID	Client Sample ID					
Method	Parameters	Result	Units	Report Limit	Analyzed	Qualifiers
50431361001	Effluent					
EPA 1633	ADONA	1.2J	ng/L	6.5	06/10/26 12:19	I
EPA 1633	NMeFOSAA	0.76J	ng/L	1.6	06/10/26 12:19	
EPA 1633	PFBS	12.2	ng/L	1.6	06/10/26 12:19	
EPA 1633	PFHxA	4.5	ng/L	1.6	06/10/26 12:19	
EPA 1633	PFBA	6.7	ng/L	6.5	06/10/26 12:19	
EPA 1633	PFMPA	1.1J	ng/L	3.3	06/10/26 12:19	
EPA 1633	PFPeA	2.4J	ng/L	3.3	06/10/26 12:19	
EPA 1633	PFHpA	1.2J	ng/L	1.6	06/10/26 12:19	
EPA 1633	PFHxS	0.71J	ng/L	1.6	06/10/26 12:19	
EPA 1633	PFNA	0.58J	ng/L	1.6	06/10/26 12:19	I
EPA 1633	PFOS	2.5	ng/L	1.6	06/10/26 12:19	
EPA 1633	PFOA	3.1	ng/L	1.6	06/10/26 12:19	

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ANALYTICAL RESULTS

Project: Effluent PFAS

Pace Project No.: 50431361

Sample: Effluent		Lab ID: 50431361001		Collected: 05/13/26 11:22		Received: 05/14/26 10:15		Matrix: Water	
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
EPA 1633 Water									
Analytical Method: EPA 1633 Preparation Method: EPA 1633									
Pace Analytical Services - Minneapolis									
11CI-PF3OUdS	<1.3	ng/L	6.5	1.3	1	06/04/26 13:44	06/10/26 12:19	763051-92-9	
3:3 FTCA	<2.1	ng/L	8.1	2.1	1	06/04/26 13:44	06/10/26 12:19	356-02-5	
4:2 FTS	<1.5	ng/L	6.5	1.5	1	06/04/26 13:44	06/10/26 12:19	757124-72-4	
5:3 FTCA	<5.5	ng/L	40.7	5.5	1	06/04/26 13:44	06/10/26 12:19	914637-49-3	
6:2 FTS	<1.5	ng/L	6.5	1.5	1	06/04/26 13:44	06/10/26 12:19	27619-97-2	
7:3 FTCA	<5.3	ng/L	40.7	5.3	1	06/04/26 13:44	06/10/26 12:19	812-70-4	
8:2 FTS	<2.5	ng/L	6.5	2.5	1	06/04/26 13:44	06/10/26 12:19	39108-34-4	
9CI-PF3ONS	<1.2	ng/L	6.5	1.2	1	06/04/26 13:44	06/10/26 12:19	756426-58-1	
ADONA	1.2J	ng/L	6.5	0.79	1	06/04/26 13:44	06/10/26 12:19	919005-14-4	I
HFPO-DA	<0.98	ng/L	6.5	0.98	1	06/04/26 13:44	06/10/26 12:19	13252-13-6	
NEtFOSAA	<0.44	ng/L	1.6	0.44	1	06/04/26 13:44	06/10/26 12:19	2991-50-6	
NEtFOSA	<0.32	ng/L	1.6	0.32	1	06/04/26 13:44	06/10/26 12:19	4151-50-2	
NEtFOSE	<3.8	ng/L	16.3	3.8	1	06/04/26 13:44	06/10/26 12:19	1691-99-2	
NFDHA	<0.82	ng/L	3.3	0.82	1	06/04/26 13:44	06/10/26 12:19	151772-58-6	
NMeFOSAA	0.76J	ng/L	1.6	0.49	1	06/04/26 13:44	06/10/26 12:19	2355-31-9	
NMeFOSA	<0.54	ng/L	1.6	0.54	1	06/04/26 13:44	06/10/26 12:19	31506-32-8	
NMeFOSE	<2.9	ng/L	16.3	2.9	1	06/04/26 13:44	06/10/26 12:19	24448-09-7	
PFBS	12.2	ng/L	1.6	0.31	1	06/04/26 13:44	06/10/26 12:19	375-73-5	
PFDA	<0.36	ng/L	1.6	0.36	1	06/04/26 13:44	06/10/26 12:19	335-76-2	
PFHxA	4.5	ng/L	1.6	0.32	1	06/04/26 13:44	06/10/26 12:19	307-24-4	
PFBA	6.7	ng/L	6.5	1.3	1	06/04/26 13:44	06/10/26 12:19	375-22-4	
PFDS	<0.48	ng/L	1.6	0.48	1	06/04/26 13:44	06/10/26 12:19	335-77-3	
PFDoS	<0.35	ng/L	1.6	0.35	1	06/04/26 13:44	06/10/26 12:19	79780-39-5	
PFEESA	<0.82	ng/L	3.3	0.82	1	06/04/26 13:44	06/10/26 12:19	113507-82-7	
PFHpS	<0.42	ng/L	1.6	0.42	1	06/04/26 13:44	06/10/26 12:19	375-92-8	
PFMBA	<0.48	ng/L	3.3	0.48	1	06/04/26 13:44	06/10/26 12:19	863090-89-5	
PFMPA	1.1J	ng/L	3.3	0.61	1	06/04/26 13:44	06/10/26 12:19	377-73-1	
PFNS	<0.45	ng/L	1.6	0.45	1	06/04/26 13:44	06/10/26 12:19	68259-12-1	
PFOSA	<0.32	ng/L	1.6	0.32	1	06/04/26 13:44	06/10/26 12:19	754-91-6	
PFPeA	2.4J	ng/L	3.3	0.45	1	06/04/26 13:44	06/10/26 12:19	2706-90-3	
PFPeS	<0.26	ng/L	1.6	0.26	1	06/04/26 13:44	06/10/26 12:19	2706-91-4	
PFDoA	<0.34	ng/L	1.6	0.34	1	06/04/26 13:44	06/10/26 12:19	307-55-1	
PFHpA	1.2J	ng/L	1.6	0.36	1	06/04/26 13:44	06/10/26 12:19	375-85-9	
PFHxS	0.71J	ng/L	1.6	0.43	1	06/04/26 13:44	06/10/26 12:19	355-46-4	
PFNA	0.58J	ng/L	1.6	0.32	1	06/04/26 13:44	06/10/26 12:19	375-95-1	I
PFOS	2.5	ng/L	1.6	0.54	1	06/04/26 13:44	06/10/26 12:19	1763-23-1	
PFOA	3.1	ng/L	1.6	0.47	1	06/04/26 13:44	06/10/26 12:19	335-67-1	
PFTeDA	<0.27	ng/L	1.6	0.27	1	06/04/26 13:44	06/10/26 12:19	376-06-7	
PFTTrDA	<0.33	ng/L	1.6	0.33	1	06/04/26 13:44	06/10/26 12:19	72629-94-8	
PFUnA	<0.35	ng/L	1.6	0.35	1	06/04/26 13:44	06/10/26 12:19	2058-94-8	
Surrogates									
13C2-PFDoA (S)	72	%	10-130		1	06/04/26 13:44	06/10/26 12:19		
13C3HFPO-DA (S)	79	%	40-130		1	06/04/26 13:44	06/10/26 12:19		
13C3-PFBS (S)	84	%	40-135		1	06/04/26 13:44	06/10/26 12:19		

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ANALYTICAL RESULTS

Project: Effluent PFAS

Pace Project No.: 50431361

Sample: Effluent		Lab ID: 50431361001		Collected: 05/13/26 11:22		Received: 05/14/26 10:15		Matrix: Water	
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
Surrogates									
13C3-PFHxS (S)	85	%.	40-130		1	06/04/26 13:44	06/10/26 12:19		
13C4-PFBA (S)	83	%.	5-130		1	06/04/26 13:44	06/10/26 12:19		
13C4-PFHpA (S)	88	%.	40-130		1	06/04/26 13:44	06/10/26 12:19		
13C5-PFHxA (S)	88	%.	40-130		1	06/04/26 13:44	06/10/26 12:19		
13C5-PFPeA (S)	81	%.	40-130		1	06/04/26 13:44	06/10/26 12:19		
13C6-PFDA (S)	78	%.	40-130		1	06/04/26 13:44	06/10/26 12:19		
13C8-PFOA (S)	85	%.	40-130		1	06/04/26 13:44	06/10/26 12:19		
13C8-PFOS (S)	77	%.	40-130		1	06/04/26 13:44	06/10/26 12:19		
13C8-PFOSA (S)	76	%.	40-130		1	06/04/26 13:44	06/10/26 12:19		
13C9-PFNA (S)	88	%.	40-130		1	06/04/26 13:44	06/10/26 12:19		
d3-MeFOSAA (S)	70	%.	40-170		1	06/04/26 13:44	06/10/26 12:19		
d3-NMeFOSA (S)	51	%.	10-130		1	06/04/26 13:44	06/10/26 12:19		
d5-EtFOSAA (S)	75	%.	25-135		1	06/04/26 13:44	06/10/26 12:19		
d5-NEtFOSA (S)	48	%.	10-130		1	06/04/26 13:44	06/10/26 12:19		
d7-NMeFOSE (S)	91	%.	10-130		1	06/04/26 13:44	06/10/26 12:19		
d9-NEtFOSE (S)	84	%.	10-130		1	06/04/26 13:44	06/10/26 12:19		
13C2-PFTA (S)	59	%.	10-130		1	06/04/26 13:44	06/10/26 12:19		
13C7-PFUdA (S)	81	%.	30-130		1	06/04/26 13:44	06/10/26 12:19		
13C24:2FTS (S)	77	%.	40-200		1	06/04/26 13:44	06/10/26 12:19		
13C26:2FTS (S)	132	%.	40-200		1	06/04/26 13:44	06/10/26 12:19		
13C28:2FTS (S)	133	%.	40-300		1	06/04/26 13:44	06/10/26 12:19		
13C3-PFPPrA (S)	41	%.	5-130		1	06/04/26 13:44	06/10/26 12:19		

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QUALITY CONTROL DATA

Project: Effluent PFAS

Pace Project No.: 50431361

QC Batch 1067853

QC Batch Method: EPA 1633

Analysis Method: EPA 1633

Analysis Description: EPA 1633 Water

Laboratory: Pace Analytical Services - Minneapolis

Associated Lab Samples: 50431361001

METHOD BLANK: 5548630

Matrix: Water

Associated Lab Samples: 50431361001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
11CI-PF3OUdS	ng/L	<1.3	6.4	1.3	06/10/26 11:38	
3:3 FTCA	ng/L	<2.1	8.0	2.1	06/10/26 11:38	
4:2 FTS	ng/L	<1.5	6.4	1.5	06/10/26 11:38	
5:3 FTCA	ng/L	<5.4	40.0	5.4	06/10/26 11:38	
6:2 FTS	ng/L	<1.5	6.4	1.5	06/10/26 11:38	
7:3 FTCA	ng/L	<5.2	40.0	5.2	06/10/26 11:38	
8:2 FTS	ng/L	<2.4	6.4	2.4	06/10/26 11:38	
9CI-PF3ONS	ng/L	<1.2	6.4	1.2	06/10/26 11:38	
ADONA	ng/L	<0.78	6.4	0.78	06/10/26 11:38	
HFPO-DA	ng/L	<0.96	6.4	0.96	06/10/26 11:38	
NETFOSA	ng/L	<0.32	1.6	0.32	06/10/26 11:38	
NETFOSAA	ng/L	<0.43	1.6	0.43	06/10/26 11:38	
NETFOSE	ng/L	<3.8	16.0	3.8	06/10/26 11:38	
NFDHA	ng/L	<0.81	3.2	0.81	06/10/26 11:38	
NMeFOSA	ng/L	<0.53	1.6	0.53	06/10/26 11:38	
NMeFOSAA	ng/L	<0.48	1.6	0.48	06/10/26 11:38	
NMeFOSE	ng/L	<2.9	16.0	2.9	06/10/26 11:38	
PFBA	ng/L	<1.3	6.4	1.3	06/10/26 11:38	
PFBS	ng/L	<0.30	1.6	0.30	06/10/26 11:38	
PFDA	ng/L	<0.36	1.6	0.36	06/10/26 11:38	
PFDaA	ng/L	<0.34	1.6	0.34	06/10/26 11:38	
PFDoS	ng/L	<0.34	1.6	0.34	06/10/26 11:38	
PFDS	ng/L	<0.47	1.6	0.47	06/10/26 11:38	
PFEESA	ng/L	<0.80	3.2	0.80	06/10/26 11:38	
PFHpA	ng/L	<0.36	1.6	0.36	06/10/26 11:38	
PFHpS	ng/L	<0.42	1.6	0.42	06/10/26 11:38	
PFHxA	ng/L	<0.31	1.6	0.31	06/10/26 11:38	
PFHxS	ng/L	<0.43	1.6	0.43	06/10/26 11:38	
PFMBA	ng/L	<0.47	3.2	0.47	06/10/26 11:38	
PFMPA	ng/L	<0.60	3.2	0.60	06/10/26 11:38	
PFNA	ng/L	<0.32	1.6	0.32	06/10/26 11:38	
PFNS	ng/L	<0.44	1.6	0.44	06/10/26 11:38	
PFOA	ng/L	<0.46	1.6	0.46	06/10/26 11:38	
PFOS	ng/L	<0.53	1.6	0.53	06/10/26 11:38	
PFOSA	ng/L	<0.31	1.6	0.31	06/10/26 11:38	
PFPeA	ng/L	<0.44	3.2	0.44	06/10/26 11:38	
PFPeS	ng/L	<0.26	1.6	0.26	06/10/26 11:38	

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QUALITY CONTROL DATA

Project: Effluent PFAS

Pace Project No.: 50431361

METHOD BLANK: 5548630

Matrix: Water

Associated Lab Samples: 50431361001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
PFTeDA	ng/L	<0.27	1.6	0.27	06/10/26 11:38	
PFTrDA	ng/L	<0.32	1.6	0.32	06/10/26 11:38	
PFUnA	ng/L	<0.35	1.6	0.35	06/10/26 11:38	
13C2-PFDoA (S)	%	76	10-130		06/10/26 11:38	
13C2-PFTA (S)	%	64	10-130		06/10/26 11:38	
13C24:2FTS (S)	%	119	40-200		06/10/26 11:38	
13C26:2FTS (S)	%	114	40-200		06/10/26 11:38	
13C28:2FTS (S)	%	116	40-300		06/10/26 11:38	
13C3-PFBS (S)	%	86	40-135		06/10/26 11:38	
13C3-PFHxS (S)	%	86	40-130		06/10/26 11:38	
13C3-PFPrA (S)	%	99	5-130		06/10/26 11:38	
13C3HFPO-DA (S)	%	94	40-130		06/10/26 11:38	
13C4-PFBA (S)	%	96	5-130		06/10/26 11:38	
13C4-PFHpA (S)	%	89	40-130		06/10/26 11:38	
13C5-PFHxA (S)	%	93	40-130		06/10/26 11:38	
13C5-PFPeA (S)	%	96	40-130		06/10/26 11:38	
13C6-PFDA (S)	%	81	40-130		06/10/26 11:38	
13C7-PFUdA (S)	%	80	30-130		06/10/26 11:38	
13C8-PFOA (S)	%	90	40-130		06/10/26 11:38	
13C8-PFOS (S)	%	81	40-130		06/10/26 11:38	
13C8-PFOSA (S)	%	81	40-130		06/10/26 11:38	
13C9-PFNA (S)	%	89	40-130		06/10/26 11:38	
d3-MeFOSAA (S)	%	89	40-170		06/10/26 11:38	
d3-NMeFOSA (S)	%	67	10-130		06/10/26 11:38	
d5-EtFOSAA (S)	%	78	25-135		06/10/26 11:38	
d5-NEtFOSA (S)	%	66	10-130		06/10/26 11:38	
d7-NMeFOSE (S)	%	85	10-130		06/10/26 11:38	
d9-NEtFOSE (S)	%	89	10-130		06/10/26 11:38	

LABORATORY CONTROL SAMPLE: 5548631

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
11Cl-PF3OUdS	ng/L	150	125	83	55-160	
3:3 FTCA	ng/L	198	151	76	65-130	
4:2 FTS	ng/L	150	139	93	70-145	
5:3 FTCA	ng/L	992	924	93	70-135	
6:2 FTS	ng/L	154	151	98	65-155	
7:3 FTCA	ng/L	992	887	89	50-145	
8:2 FTS	ng/L	154	144	93	60-150	
9Cl-PF3ONS	ng/L	150	135	90	70-155	
ADONA	ng/L	150	136	91	65-145	
HFPO-DA	ng/L	160	147	92	70-140	

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Date: 06/12/2026 12:00 AM



QUALITY CONTROL DATA

Project: Effluent PFAS

Pace Project No.: 50431361

LABORATORY CONTROL SAMPLE: 5548631

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
NEtFOSA	ng/L	38.4	36.1	94	65-145	
NEtFOSAA	ng/L	38.4	35.4	92	70-145	
NEtFOSE	ng/L	384	369	96	70-135	
NFDHA	ng/L	80	69.7	87	50-150	
NMeFOSA	ng/L	38.4	35.2	92	60-150	
NMeFOSAA	ng/L	38.4	35.8	93	50-140	
NMeFOSE	ng/L	384	385	100	70-145	
PFBA	ng/L	160	152	95	70-140	
PFBS	ng/L	35.2	33.3	95	60-145	
PFDA	ng/L	38.4	37.9	99	70-140	
PFDaA	ng/L	38.4	37.4	97	70-140	
PFDoS	ng/L	38.4	29.2	76	50-145	
PFDS	ng/L	38.4	31.7	83	60-145	
PFEESA	ng/L	70.4	59.2	84	70-140	
PFHpA	ng/L	38.4	36.1	94	70-150	
PFHpS	ng/L	38.4	36.5	95	70-150	
PFHxA	ng/L	38.4	36.9	96	70-145	
PFHxS	ng/L	35.2	33.3	95	65-145	
PFMBA	ng/L	80	75.0	94	60-150	
PFMPA	ng/L	80	77.5	97	55-140	
PFNA	ng/L	38.4	36.6	95	70-150	
PFNS	ng/L	38.4	38.1	99	65-145	
PFOA	ng/L	38.4	37.9	99	70-150	
PFOS	ng/L	38.4	33.9	88	55-150	
PFOSA	ng/L	38.4	36.0	94	70-145	
PFPeA	ng/L	80	73.5	92	65-135	
PFPeS	ng/L	38.4	35.3	92	65-140	
PFTeDA	ng/L	38.4	38.8	101	60-140	
PFTrDA	ng/L	38.4	33.5	87	65-140	
PFUnA	ng/L	38.4	36.7	95	70-145	
13C2-PFDaA (S)	%			88	10-130	
13C2-PFTA (S)	%			80	10-130	
13C24:2FTS (S)	%			129	40-200	
13C26:2FTS (S)	%			120	40-200	
13C28:2FTS (S)	%			138	40-300	
13C3-PFBS (S)	%			106	40-135	
13C3-PFHxS (S)	%			105	40-130	
13C3-PFPrA (S)	%			111	5-130	
13C3HFPO-DA (S)	%			111	40-130	
13C4-PFBA (S)	%			110	5-130	
13C4-PFHpA (S)	%			111	40-130	
13C5-PFHxA (S)	%			113	40-130	
13C5-PFPeA (S)	%			114	40-130	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Effluent PFAS
Pace Project No.: 50431361

LABORATORY CONTROL SAMPLE: 5548631

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
13C6-PFDA (S)	%.			103	40-130	
13C7-PFUdA (S)	%.			100	30-130	
13C8-PFOA (S)	%.			97	40-130	
13C8-PFOS (S)	%.			103	40-130	
13C8-PFOSA (S)	%.			98	40-130	
13C9-PFNA (S)	%.			106	40-130	
d3-MeFOSAA (S)	%.			106	40-170	
d3-NMeFOSA (S)	%.			85	10-130	
d5-EtFOSAA (S)	%.			98	25-135	
d5-NEtFOSA (S)	%.			84	10-130	
d7-NMeFOSE (S)	%.			96	10-130	
d9-NEtFOSE (S)	%.			102	10-130	

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QUALIFIERS

Project: Effluent PFAS

Pace Project No.: 50431361

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.

ND - Not Detected at or above adjusted reporting limit.

TNTC - Too Numerous To Count

J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.

MDL - Adjusted Method Detection Limit.

PQL - Practical Quantitation Limit.

RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.

S - Surrogate

1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.

Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.

LCS(D) - Laboratory Control Sample (Duplicate)

MS(D) - Matrix Spike (Duplicate)

DUP - Sample Duplicate

RPD - Relative Percent Difference

NC - Not Calculable.

SG - Silica Gel - Clean-Up

U - Indicates the compound was analyzed for, but not detected.

N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.

Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.

Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.

TNI - The NELAC Institute.

ANALYTE QUALIFIERS

I The ion ratio is outside control limits.

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Effluent PFAS
Pace Project No.: 50431361

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50431361001	Effluent	EPA 1633	1067853	EPA 1633	1069041



REPORT OF LABORATORY ANALYSIS

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CHAIN-OF-CUSTODY Analytical Request Document

Chain-of-Custody is a LEGAL DOCUMENT - Complete all relevant fields

Company Name:	City of Cadillac	Contact/Report To:	Cindy Tomaszewski
Street Address:	1121 Plett Rd Cadillac, MI 49601	Phone #:	231-775-2368
		E-Mail:	ctomaszewski@cadillac-mi.net
		Cc E-Mail:	
Customer Project #:		Invoice To:	Accounts Payable
Project Name:	Effluent PFAS	Invoice E-Mail:	ctomaszewski@cadillac-mi.net
		Purchase Order # (if applicable):	
		Quote #:	

Time Zone Collected: [] AK [] PT [] MT [] CT [X] ET [] Michigan									
Data Deliverables:									
Regulatory Program (DW, RCRA, etc.) as applicable: Reportable [] Yes [] No									
NPDES Rush (Pre-approval required):									
[] Level II [] Level III [] Level IV [] Same Day [] 1 Day [] 2 Day [] 3 Day [] Other									
[] JEQUIS DW PWSID # or WW Permit # as applicable:									
Date Results Requested: Field Filtered (if applicable): [] Yes [X] No									
[] Other Analysis:									

[illegible]

Additional Instructions from Pace®:	Collected By: (Printed Name) <i>Taysha Laezko</i>	Received by/Company: (Signature) <i>T. Laezko</i>	Received by/Company: (Signature) <i>Ship via UPS Ground</i>
	Signature: <i>T. Laezko</i>		Received by/Company: (Signature)
	Date/Time: <i>5-13-26 / 1135</i>		Received by/Company: (Signature)
	Date/Time:		Received by/Company: (Signature)
	Date/Time:		Received by/Company: (Signature)
	Date/Time:		Received by/Company: (Signature)

[illegible]

Customer Remarks / Special Conditions / Possible Hazards:			
# Coolers:	Thermometer ID:	Correction Factor (°C):	Obs. Temp. (°C)
			Corrected Temp. (°C)
			On Ice:
Tracking Number:			
Date/Time:	Delivered by: [] In- Person [] Courier		
Date/Time:	[] FedEx [] UPS [] Other		
Date/Time:	Page: 1 of 1		



Sample Conditions Upon Receipt Form (SCUR)

Date/Time: <u>5.14.26</u>	Evaluated By: <u>WDC</u>	WO#: 50431361	
Client: <u>City of Cadillac</u>	PM: <u>BJH</u>	PM: <u>BJH</u>	Due Date: <u>06/08/26</u>
Lab Notified of Rush or Short Holds: <u>YES</u> NO NA		CLIENT: <u>GR-Cadillac</u>	
Project Received Via: <u>FedEx</u> UPS Client Pace Courier Other: _____			Comments:
Custody Seal Present and Intact:	<u>YES</u>	NO	NA
Received Sample Information Form (SIF): Drinking Waters Only	<u>YES</u>	NO	NA
Short Hold Present (≤ 48 Hours):	<u>YES</u>	NO	NA
Sample Received in Hold:	<u>YES</u>	NO	NA
Custody Signature Present:	<u>YES</u>	NO	NA
Collector Signature Present:	<u>YES</u>	NO	NA
Sample Collected Today and On Ice:	<u>YES</u>	NO	NA
IR Gun #: <u>350</u> 351 Therm #: <u>282</u> <u>283</u>	Temp. should be 0°C - 6°C (Initial/Corrected)		
Ice Type: <u>WET Bagged / WET Loose</u> BLUE NONE	1. Cooler Temp. Upon Receipt: <u>3.9/3.8</u> °C		
Ice Location: <u>TOP</u> BOTTOM MIDDLE DISPERSED	2. Cooler Temp. Upon Receipt: _____ °C		
Temp Blank Received:	<u>YES</u>	NO	NA
Sample Label Matches COC (ID/Date/Time):	<u>YES</u>	NO	NA
Container Intact:	<u>YES</u>	NO	NA
Correct Container:	<u>YES</u>	NO	NA
Sufficient Volume:	<u>YES</u>	NO	NA
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation. Denote with pink sticker on the lid if unacceptable. pH Strip Lot #: _____ <i>Exceptions are VOA, coliform, LLHg, O&G/TPH, or any container with a septum cap or preserved with HCl</i>	<u>YES</u>	NO	NA
Residual Chlorine Absent: <u>Cl₂ Strip Lot #: _____</u> <i>Applies to SVOC 625, PCB/Pest. 608, Total/Amenable/Available/Free Cyanide</i>	<u>YES</u>	NO	NA
VOA Headspace Acceptable (<6mm): Denote with silver x on rim of container cap if unacceptable.	<u>YES</u>	NO	NA
Trip Blank Received: <u>HCl</u> MeOH Other: _____	<u>YES</u>	NO	<u>ON HOLD</u>
Comments:	3. Cooler Temp. Upon Receipt: _____ °C		
	4. Cooler Temp. Upon Receipt: _____ °C		
	Non-Conformance Form Required: <u>YES</u> NO		